# Your Space We Care

10年质保+图标
+放大

# Smart Film

**Filmbase HQ:**
31F, 3A Building,Smart Park,Baolong street, Longgang district ,Shenzhen ,China

**Filmbase Branch:**
71F ,PingAn international Center, FuTian district, Shenzhen, China

**R&D Center:**
Building3#, Huaqiang industry Park. Baolong street. Longgang district. Shenzhen,China

**Filmbase Manufacturing Center :**
Building2+Building3, Huagiang Industry Park. Baolong street. Longgang district. Shenzhen, China

**Filmbase ITO Factory:**
1st Floor, Hongxing Industrial Park, Lilin Village, Lilin Town, Zhongkai High-tech Zone, Huizhou City ,Guangdong,China

**Chongqing Factory:**
Factory Building#7, No.3 Jjianyuan Road, Tiao Deng Town, Dadukou District, Chongqing,China

**Filmbase Tel:**
+86 755-28505346
+86 139 28440802
+86 1912930 8937

**Filmbase Email:**
info@smartfilmbase.com

**Filmbase Web:**
www.smartfilmbase.com

**Filmbase USA :**
1053 NE 43rd St, Oakland Park ,FL 33334 , Florida , USA
Tel: +1（754）333-0959
Email：usa.james@filmbase.cn

**Filmbase Korea :**
#207 Bu Pyeong Dae-ro 337,Bu Pyeong Gu, In cheon si,south Korea
Tel：+82 10 9029 3653
Email：korea@smartfilmbase.com

所有认证 横向排版

100多项专利



FILMBASE®

删除

# Smart Film

## Protect privacy,
## reshape spatial experience,
## make buildings smarter.

At Filmbase, we believe smart materials should deliver more than on-demand opacity. They should enable new ways to design, manage, and experience space–bringing together privacy, comfort, safety, and interactive capability in one solution. Our PDLC Smart Film turns ordinary glass into a responsive architectural surface, adapting instantly to people and the surrounding environment.

# PDLC Evolution

## Where the Industry Came From—and Where It's Going

PDLC (Polymer Dispersed Liquid Crystal) originated in the United States, where early breakthroughs were protected by foundational patents and brought to market through initial commercialization efforts. Over time, it transitioned from a laboratory innovation into a proven industrial technology, as leading global players licensed, adopted, and scaled PDLC for real-world applications.

### The 20th century

The 1980s-1990s defined PDLC's first commercialization wave and, critically, its IP structure. Two major routes matured in parallel: NCAP (microemulsion)—invented and patented by J. L. Fergason in 1984, exclusively licensed by Raychem in 1987, and first commercialized by Raychem's subsidiary Taliq—and PDLC (phase separation)—patented and trademarked by Kent State University (KSU) in 1987 and licensed to global industrial players. This era validated market demand but was constrained by high cost and uneven reliability, and it was shaped by cross-region patent conflicts that limited scale in parts of Japan and Europe.

### The 21th century

A major turning point came in the early 2000s, when key patent barriers fell away—Raychem's PDLC-related patents expired in 2002, followed by KSU's patents in 2005. The expiration of these foundational patents materially reduced entry barriers, catalyzing global manufacturer expansion and accelerating adoption. The technology became significantly more accessible. Importantly, this was not just a volume story—it coincided with manufacturing and materials upgrades that improved economics and performance: broader use of UV curing, movement toward solvent-free processes, increasing automation, and improved LC/polymer formulations (e.g., lower switching thresholds, higher contrast, better durability), collectively driving better yields and more bankable product performance.

更多关于产品信息 放在后面

### In 2017

With the market opening up, Filmbase emerged to accelerate the next wave of PDLC innovation—strengthening supply chains, scaling high-quality production, and rapidly expanding PDLC applications across regions and industries. In 2017, our founder, Zhu Longshan, introduced revolutionary innovations and process reforms built on the Raychem and KSU technology foundations, significantly improving both output capacity and product quality. Mr. Zhu established Filmbase's end-to-end technical stack across the critical layers of PDLC manufacturing—from ITO and PET substrates, adhesives, and liquid crystal systems to coating processes and proprietary, patented material formulations. Building on this platform, Filmbase has developed 12 distinct smart film series, covering a wide range of transparency and haze levels, colors, functions, solution configurations, and maximum size formats, enabling partners to serve diverse architectural, commercial, and mobility applications with precision.

PDLC is in a strong expansion cycle.Intensifying competition, lower production costs and pricing, improved glass technologies, and broader market education have significantly accelerated demand. The smart dimming materials market is described as growing rapidly, with expectations to reach a multi-billion-dollar scale by 2031. Regional adoption remains differentiated—North America and Europe show strong demand, while parts of Asia still rely more heavily on traditional partition solutions.

Future growth will be driven by PDLC's evolution from a "privacy switch" into a foundational smart material for buildings and mobility—accelerated by smart home integration, energy-efficient commercial space upgrades, and wider adoption across architecture and automotive applications, as global priorities shift toward sustainability and energy saving.

# Why Filmbase Smart Film?

## Our Innovation Power New Material Technology

移动到后面 more

<1>

### Innovation

In 2017. Filmbase introduced major innovations and manufacturing reforms based on established PDLC technology routes, significantly improving both output and product quality.

<2>

### The largest manufacturer

By 2023, Filmbase is described as the world's largest PDLC film manufacturer.

<3>

### Master the supply chain

Filmbase is described as the only company covering R&D and production across key core raw materials–including ITO, PET, and glue–allowing us to control quality consistency, cost stability, and lead time from the source.

<4>

### 40% of profits are invested in research and development.

We reinvest 40% of annual profits into new technologies and new products to sustain long-term leadership and deliver continuously improved solutions for partners.

<5>

### 105 countries and regions worldwide

Our products are exported to 105 countries and regions, with strategic partnerships across the global PDLC ecosystem.

<6>

### Exclusive market protection

"market protection" with one exclusive agent per country/region.

# Why Filmbase Smart Film?

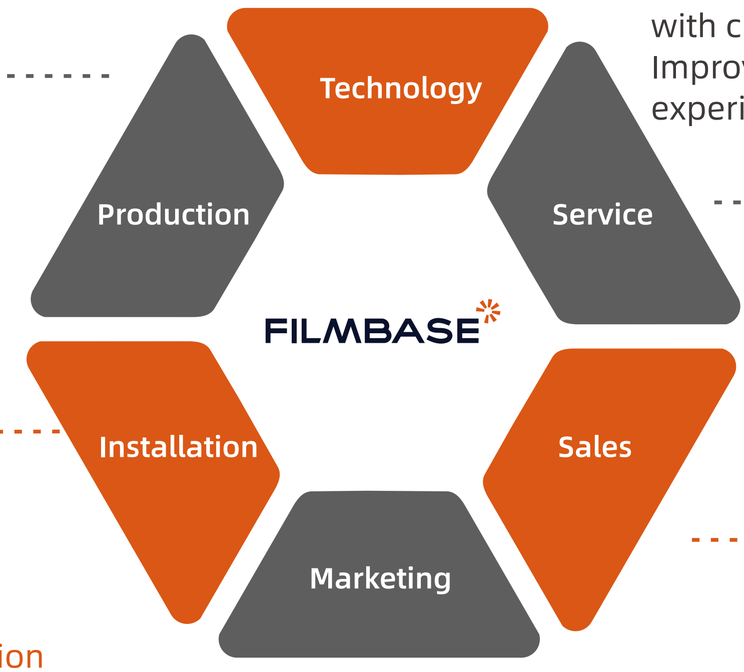
## To accelerate partner success, we provide comprehensive training

调到后面

Local processing
and production
Fast delivery

Latest technology
Latest business
opportunities

How to follow up
with customers
Improve customer
experience




Build the installation
team，Receive support
from the headquarters
Training schedule

Improve sales skills
Achieve outstanding performance

Build the installation
team，Receive support
from the headquarters
Training schedule

删除

# GLOBAL CERTIFICATION

Hold proprietary patents

PS E CE

CB NOM RoHS

FC IATF16949 ISO 9001 ISO 14001

ISO 45001

调整到后面

# Market Prospects

From 2026 to 2036, the global dimmable film market will experience steady growth, expanding from approximately $2.1 billion to over $4 billion. Regional development patterns will exhibit distinct characteristics, with the Asia-Pacific region emerging as the core growth engine.

The Asia-Pacific region is the largest and fastest-growing market for dimmable films globally, projected to reach $1.6–1.8 billion by 2036 with a CAGR of 12–15%. Key drivers include advancing urbanization, implementation of green building policies, and surging demand for new energy vehicles. North America and Europe will see steady growth but a slight decline in market share, reaching $1.0–1.2 billion and $0.8–0.92 billion respectively by 2036, with CAGR of 8%–10% and 10%–12%. Implementation of energy efficiency regulations and upgrades to automotive smart cockpits will be the primary growth drivers.

Latin America, the Middle East, and Africa are experiencing modest growth, with market sizes projected to reach $240 million to $320 million and $160 million to $240 million respectively by 2036. Regional economic development and infrastructure investments continue to drive demand. Overall, factors such as declining technology costs, expanding application scenarios, and intensifying market competition will collectively shape the long-term growth trajectory of the global dimming film market.

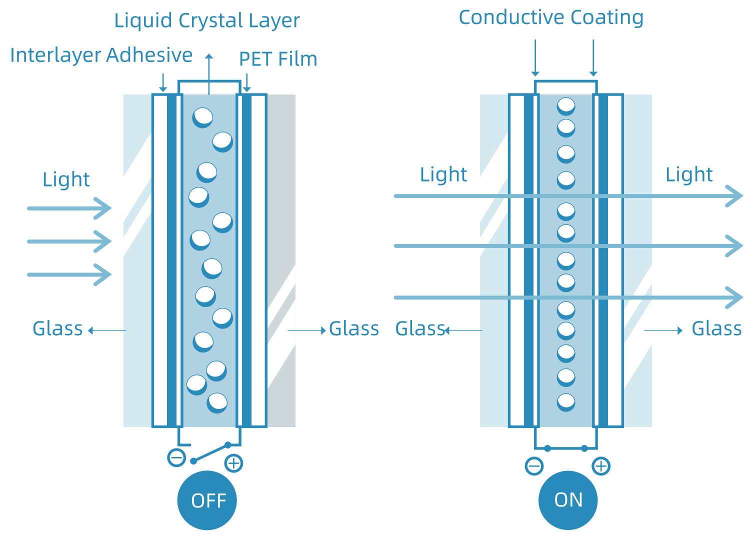
这个页面调整到第三页

# Smart Film Technology Principles

Smart film is a new type of electronic light control product. The electronically controlled smart film is a liquid crystal/polymer hybrid material injected in the middle of two transparent conductive films.

PDLC film in the absence of an electric field, the electronically controlled smart film is in an opaque state.

When an alternating current is applied, the liquid crystal molecules are arranged in an orderly manner, at which point the electrically controlled dimming film changes from an opaque state (OFF) to a transparent state (ON).

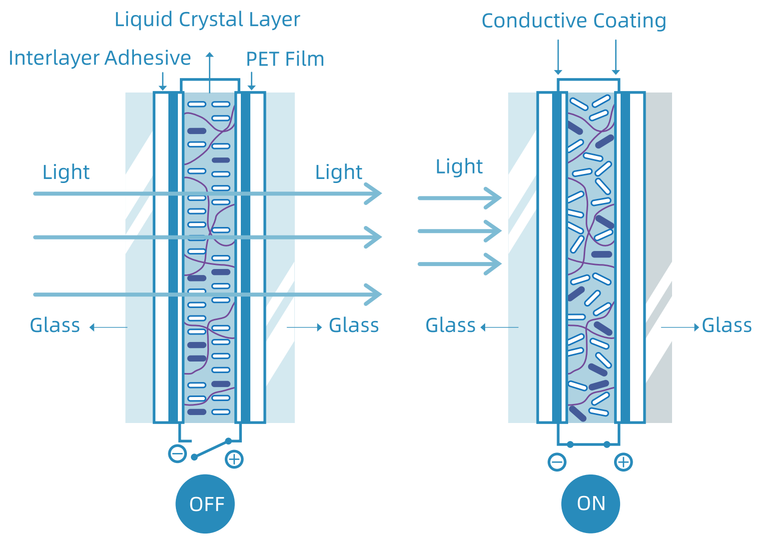



PDLC

PNLC film different from normal PDLC smart film,the working principle of PNLC film is opaque-power on,clear power off.

The electric field enables a rapid transition from the ON state to the OFF state and from the OFF state to the ON state.

Smart film is usually applied to the surface of the glass or laminated between the glass layers, making the otherwise transparent glass, more possible.

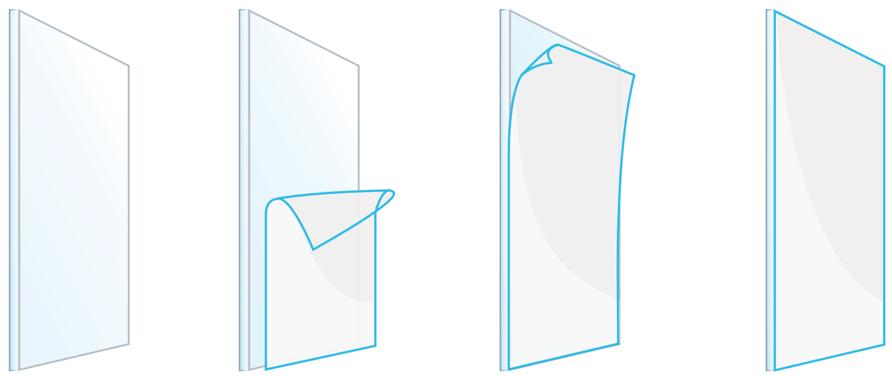



PNLC

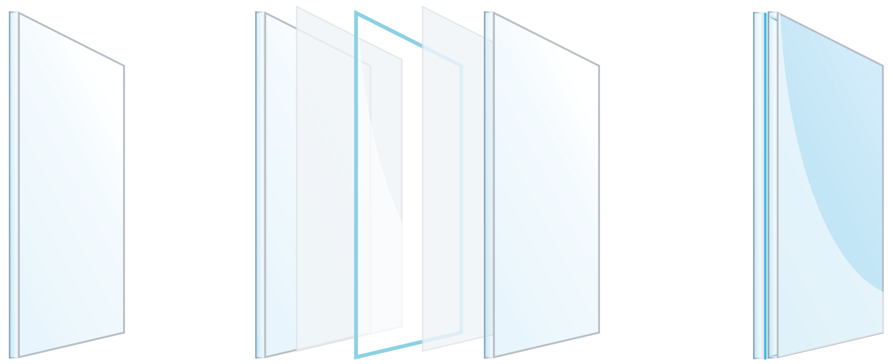
# Two types

## Smart Film

There are two types of Smart films, Self-adhesive Smart film and Laminated Smart film.



Self-adhesive Smart film

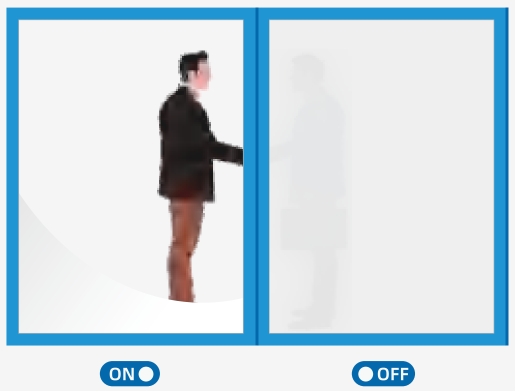


Laminated Smart film

调整到第二页

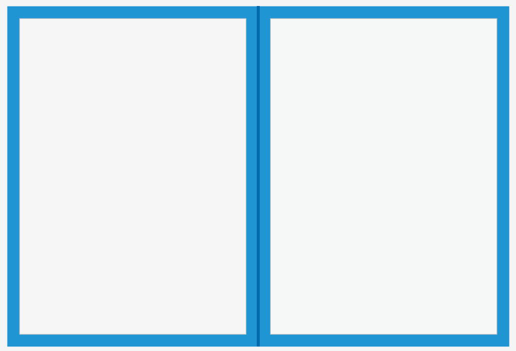
# Characteristics

## Smart Film



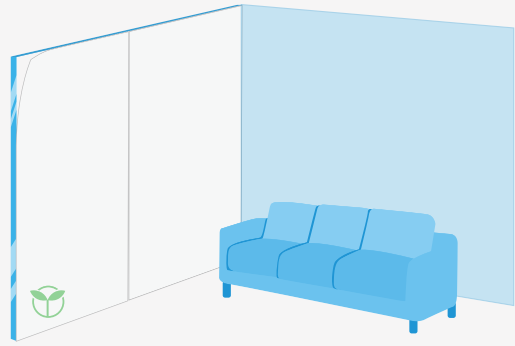

### Privacy protection

To protect your privacy and security, you can control it to be transparent or opaque at any time. Make your space more malleable, choose open or private at will.



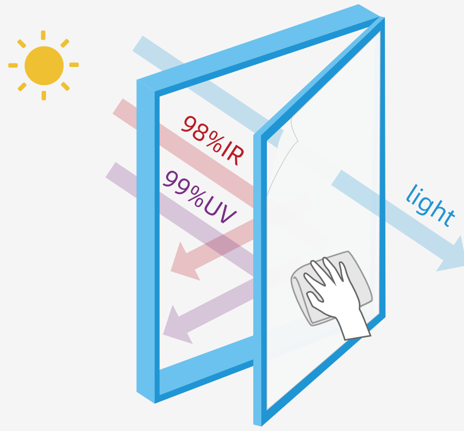
### Security

Even if the glass is broken, the fragments can still be adsorbed on the PDLC film, which can avoid personal injury caused by broken glass.



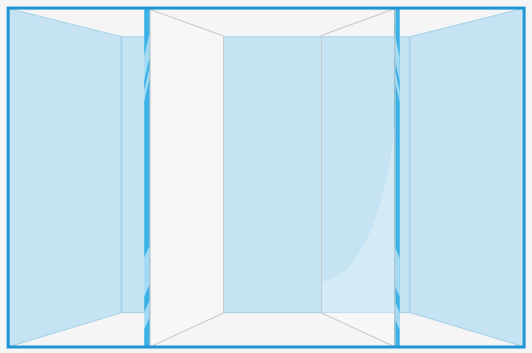
### Environmental protection

The installation of PDLC film can effectively reduce the loss of decoration, which can better reduce pollution and protect the environment.



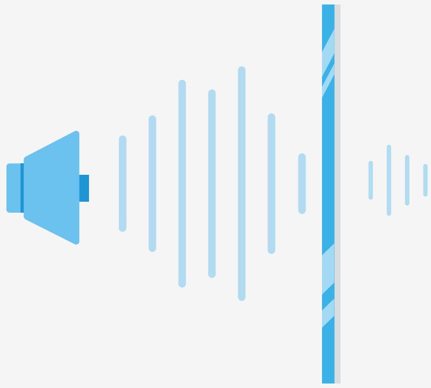

### Heat-resistant and anti-scratch

The maximum operating temperature is 105℃, which allows the product to continue to operate at higher temperatures. The anti-scratch layer protects the surface of the product from daily scratches and does not affect the clarity of the product itself.



### Space partition

With the use of PDLC film, glass partitions can easily differentiate between different spaces, making your three-dimensional space more malleable and versatile.

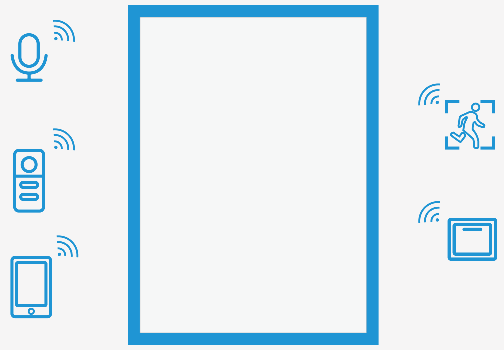


### Anti-noise

Compared with ordinary glass, the anti-noise ability of glass using Filmbase PDLC film is increased by 20%.

### Touch screen and projection

Using a high-definition projector, it is easy to realize the viewing scene, with a viewing angle of 180°, 4K picture quality display effect, and with the radar can be multi-touch, video interaction.



### Various control methods

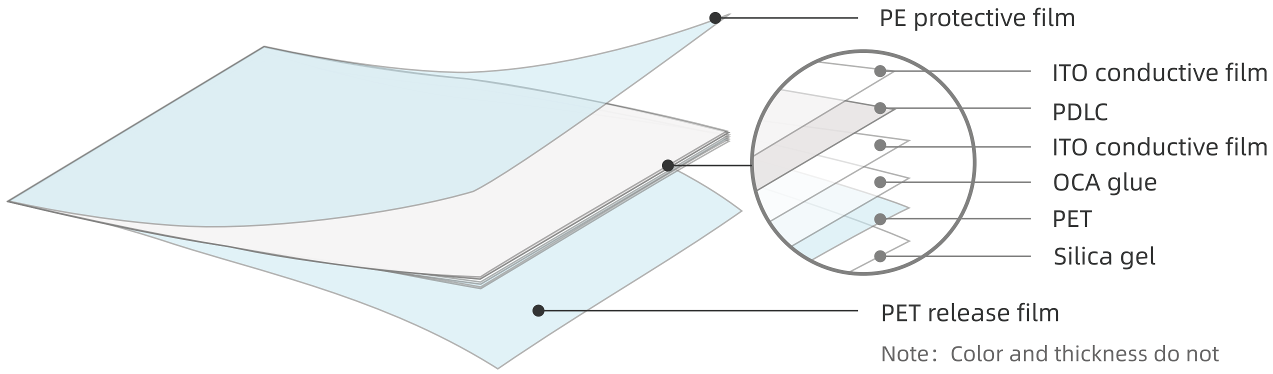
In order to meet your various needs, you can choose your preferred control method: remote control, wired control, infrared induction, voice control, mobile application, etc. In addition, an FM transformer can be selected to gradually change the film.


# FILMBASE 091

FILMBASE 091 is the luxury PDLC film in the today market with 90%-92% transmittance,< 3% haze, which maximizes the clarity of the original glass and enhances privacy.

It is used in scientific research institutions, intelligence agencies, luxury villas, president's office, super 5-star hotels, Hollywood theaters, and other places that require ultra-transparency and absolute privacy.

增加夹胶系列 产品图示




Note：Color and thickness do not represent the actual situation, only for the convenience of illustration.

## Size

FILMBASE 091

| Items | | | |
|---|---|---|---|
| Adhesive PDLC film | Total Thickness | | 0.47mm |
| | Cut to Sheet | Width | 100-1800(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1200,1500,1800(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |
| Laminated PDLC film | Total Thickness | | 0.38mm |
| | Cut to Sheet | Width | 100-1800(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1200,1500,1800(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |

## Optical Parameter

FILMBASE 091

| Items | ON/OFF | | Test Methods/Standards |
|---|---|---|---|
| Operating voltage | ON | 60V | Multimeter/ Voltage Regulator,WGW |
| Power consumption | ON | ≤ 5W/m² | Multi-parameter electric measuring instrument |
| Viewing angle | ON | ≥ 160° | Visual Inspection |
| Uv blocking rate | OFF | ≥ 99% | Optical transmittance tester |
| Infrared blocking rate | OFF | ≥ 60% | |
| Response time | OFF→ON | ≤ 30 ms | Liquid crystal parameter integrated tester |
| | ON→OFF | ≤ 200 ms | |
| Operating temperature | -- | -30℃ ~ 50℃ | constant temperature and humidity testing machine |
| Life-span | -- | 100000 Hours | GB18910.5-2008 |
| Switching times | -- | ≥2 million times | Self-made on/off tester |

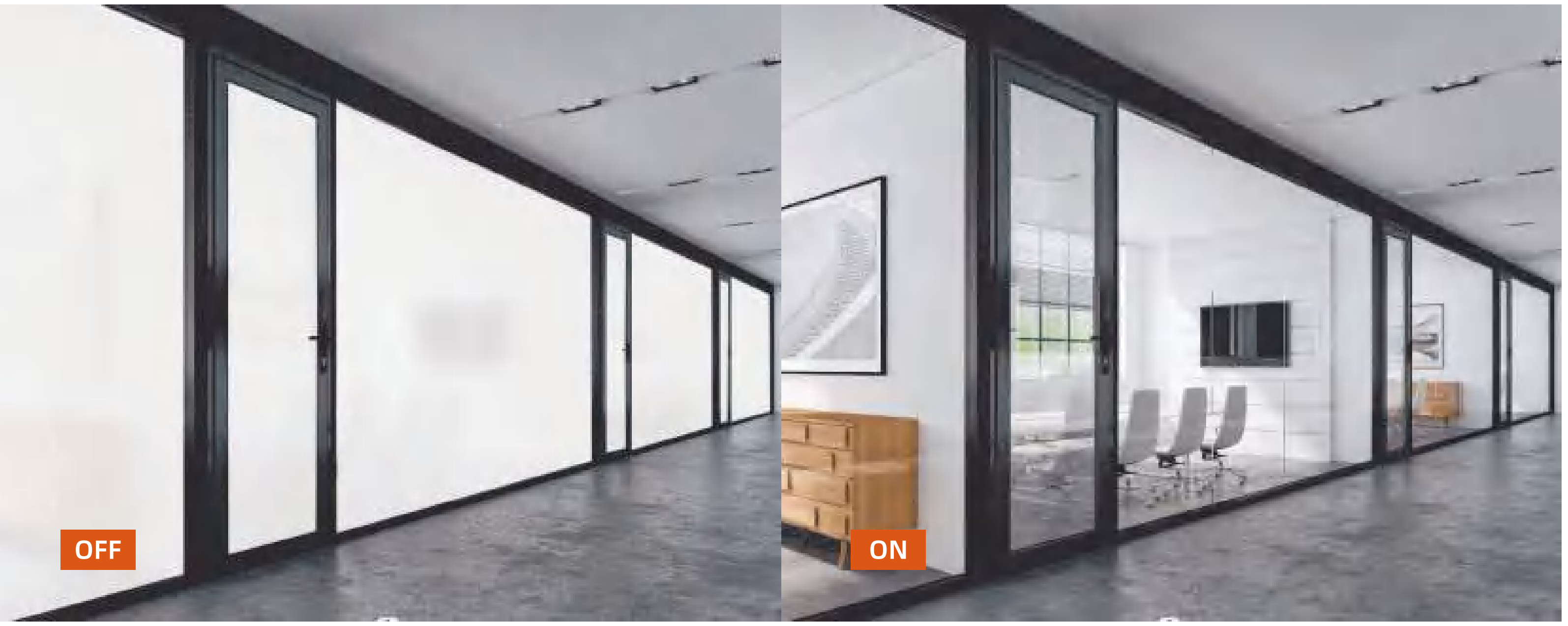
PDLC film is driven by AC power, colorless and transparent when power is added, and fogged and translucent when power is removed, and its photoelectric performance is shown in the table above.

## Product Transparency / Haze

FILMBASE 091

| Items | Full light tester | | | | Parallel light tester | |
|---|---|---|---|---|---|---|
| | ON/OFF | Haze | Full light transmittance | b value | 45° light | Parallel light |
| Adhesive PDLC film | OFF | 94.5 | 85.0 | 7.57 | / | / |
| | ON | 3.00 | 91.0 | 5.29 | 60.0 | 88.0 |
| Laminated PDLC film | OFF | 94.5 | 85.0 | 6.79 | / | / |
| | ON | 3.00 | 91.0 | 4.31 | 60.0 | 88.0 |

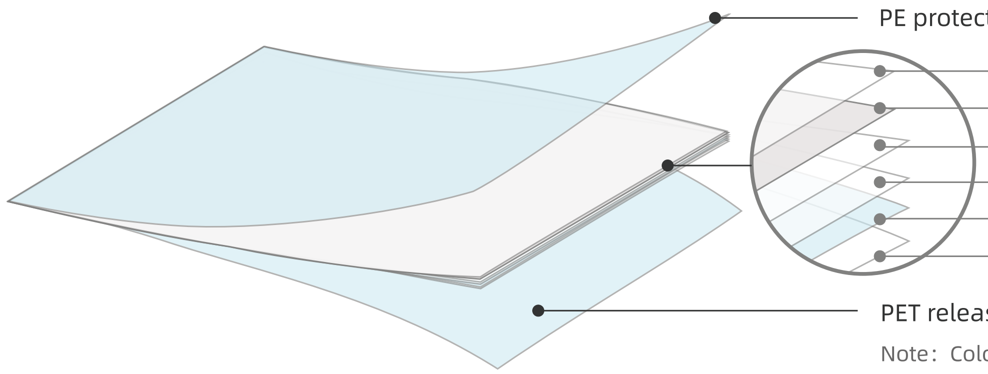
FILMBASE | SMART FILM




# FILMBASE 087

087 is Filmbase most hot sales item ,Super clear 87-89% Transmittance and nice 3-4% haze .

It is suitable for modern office partitions, space partitions, conference halls, museums, exhibition centers, star hotels, bars, shops, hospitals and other places that need to integrate the functions of transparency, privacy, partition and projection display.




Note: Color and thickness do not represent the actual situation, only for the convenience of illustration.

## Size

FILMBASE 087

| Items | | | |
|---|---|---|---|
| Adhesive PDLC film | Total Thickness | | 0.47mm |
| | Cut to Sheet | Width | 100-1800(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1000,1200,1500,1800(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |
| Laminated PDLC film | Total Thickness | | 0.38mm |
| | Cut to Sheet | Width | 100-2100(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1000,1200,1400,1500,1800,2000,2100(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |

## Optical Parameter

FILMBASE 087

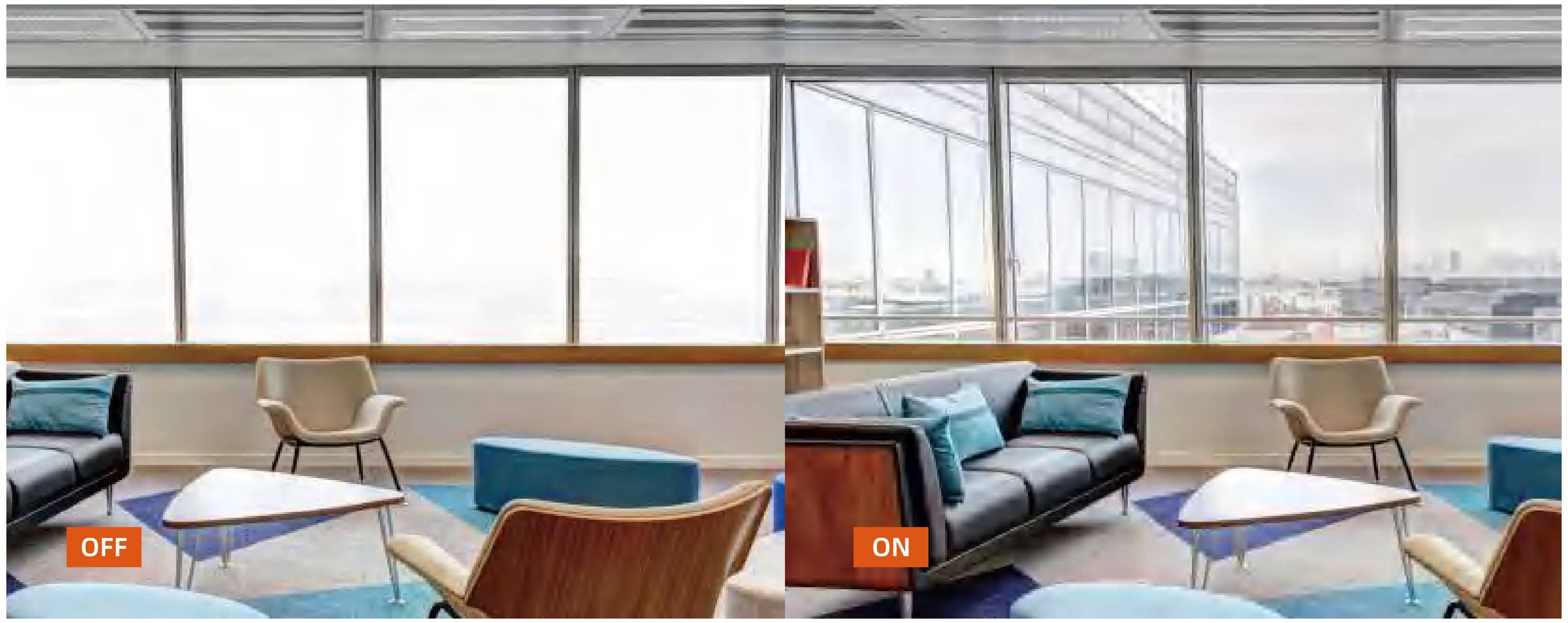
| Items | ON/OFF | | Test Methods/Standards |
|---|---|---|---|
| Operating voltage | ON | 60V | Multimeter/ Voltage Regulator,WGW |
| Power consumption | ON | ≤ 5W/m² | Multi-parameter electric measuring instrument |
| Viewing angle | ON | ≥ 160° | Visual Inspection |
| Uv blocking rate | OFF | ≥ 99% | Optical transmittance tester |
| Infrared blocking rate | OFF | ≥ 60% | |
| Response time | OFF→ON | ≤ 30 ms | Liquid crystal parameter integrated tester |
| | ON→OFF | ≤ 200 ms | |
| Operating temperature | -- | -30°C ~ 50°C | constant temperature and humidity testing machine |
| Life-span | -- | 100000 Hours | GB18910.5-2008 |
| Switching times | -- | ≥2 million times | Self-made on/off tester |

PDLC film is driven by AC power, colorless and transparent when power is added, and fogged and translucent when power is removed, and its photoelectric performance is shown in the table above.

## Product Transparency / Haze

FILMBASE 087

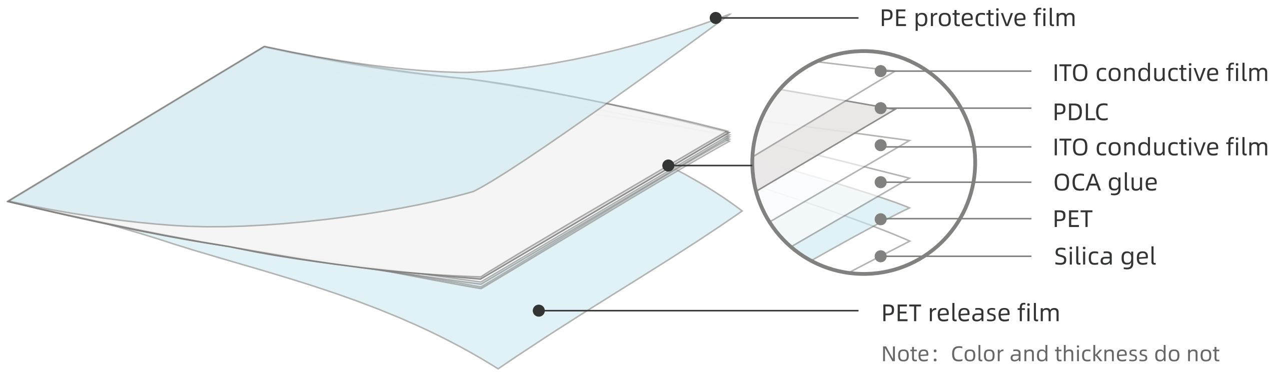
| Items | Full light tester | | | | Parallel light tester | |
|---|---|---|---|---|---|---|
| | ON/OFF | Haze | Full light transmittance | b value | 45° light | Parallel light |
| Adhesive PDLC film | OFF | 94.5 | 78.0 | 6.95 | / | / |
| | ON | 4.53 | 88.0 | 4.71 | 58.0 | 86.0 |
| Laminated PDLC film | OFF | 94.5 | 78.0 | 5.9 | / | / |
| | ON | 4.53 | 88.0 | 3.72 | 58.0 | 86.0 |




# FILMBASE 085

085 is Filmbase most cost effective item,Nice clear 84-86% Transmittance and nice 4-5% haze ,which is very welcome by the projects which does not request super high transparency and low haze applications.

It is suitable for modern office partitions, space partitions, conference halls, museums, exhibition centers, star hotels, bars, shops, hospitals and other places that need to integrate the functions of transparency, privacy, partition and projection display.




Note: Color and thickness do not represent the actual situation, only for the convenience of illustration.

## Size

FILMBASE 085

| Items | | | |
|---|---|---|---|
| Adhesive PDLC film | Total Thickness | | 0.47mm |
| | Cut to Sheet | Width | 100-1800(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1000,1200,1500,1800(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |
| Laminated PDLC film | Total Thickness | | 0.38mm |
| | Cut to Sheet | Width | 100-2100(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1000,1200,1500,1800,2000,2100(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |

## Optical Parameter

FILMBASE 085

| Items | ON/OFF | | Test Methods/Standards |
|---|---|---|---|
| Operating voltage | ON | 60V | Multimeter/ Voltage Regulator,WGW |
| Power consumption | ON | ≤ 5W/m² | Multi-parameter electric measuring instrument |
| Viewing angle | ON | ≥ 160° | Visual Inspection |
| Uv blocking rate | OFF | ≥ 99% | Optical transmittance tester |
| Infrared blocking rate | OFF | ≥ 60% | |
| Response time | OFF→ON | ≤ 30 ms | Liquid crystal parameter integrated tester |
| | ON→OFF | ≤ 200 ms | |
| Operating temperature | -- | -30°C ~ 50°C | constant temperature and humidity testing machine |
| Life-span | -- | 100000 Hours | GB18910.5-2008 |
| Switching times | -- | ≥2 million times | Self-made on/off tester |

PDLC film is driven by AC power, colorless and transparent when power is added, and fogged and translucent when power is removed, and its photoelectric performance is shown in the table above.

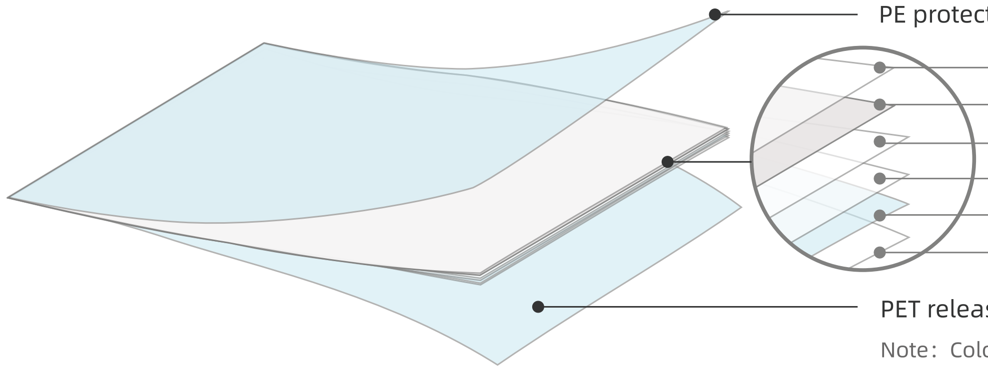
## Product Transparency / Haze

FILMBASE 085

| Items | Full light tester | | | | Parallel light tester | |
|---|---|---|---|---|---|---|
| | ON/OFF | Haze | Full light transmittance | b value | 45° light | Parallel light |
| Adhesive PDLC film | OFF | 94.5 | 78.0 | 6.62 | / | / |
| | ON | 4.53 | 84.0 | 4.56 | 58.0 | 82.0 |
| Laminated PDLC film | OFF | 94.5 | 78.0 | 5.64 | / | / |
| | ON | 4.53 | 84.0 | 3.55 | 58.0 | 81.0 |

# FILMBASE Warrior

Filmbase Warrior series PDLC film is designed for high temperature application, Such as large-scale architectural curtain walls, outdoor partitions, glass balconies, outdoor swimming pool fences etc.




Note: Color and thickness do not represent the actual situation, only for the convenience of illustration.

## Size

FILMBASE Warrior

| Items | | | |
|---|---|---|---|
| Adhesive PDLC film | Total Thickness | | 0.47mm |
| | Cut to Sheet | Width | 100-1500(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1200,1500(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |

## Optical Parameter

FILMBASE Warrior

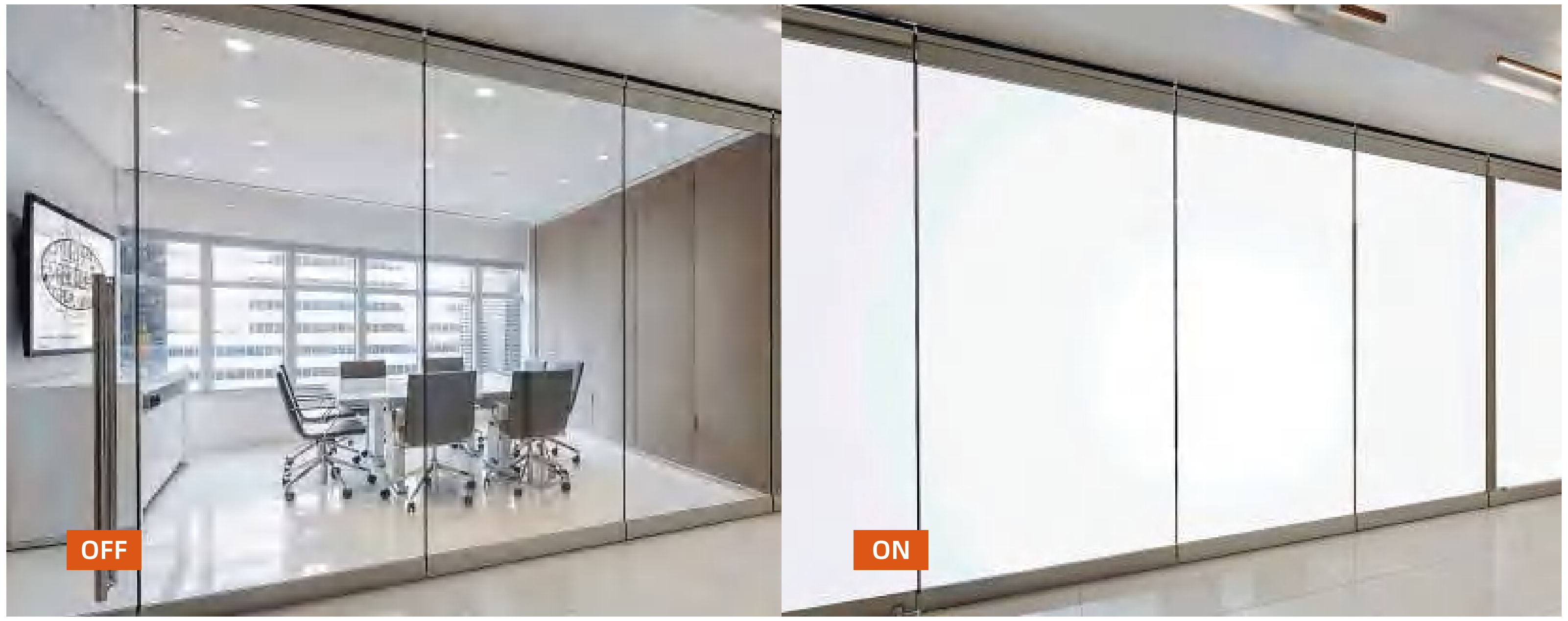
| Items | ON/OFF | | Test Methods/Standards |
|---|---|---|---|
| Operating voltage | ON | 60V | Multimeter/ Voltage Regulator,WGW |
| Power consumption | ON | ≤ 5W/m² | Multi-parameter electric measuring instrument |
| Viewing angle | ON | ≥ 160° | Visual Inspection |
| Uv blocking rate | OFF | ≥ 99% | Optical transmittance tester |
| Infrared blocking rate | OFF | ≥ 60% | |
| Response time | OFF→ON | ≤ 30 ms | Liquid crystal parameter integrated tester |
| | ON→OFF | ≤ 200 ms | |
| Operating temperature | -- | -30℃ ~ 90℃ | constant temperature and humidity testing machine |
| Life-span | -- | 100000 Hours | GB18910.5-2008 |
| Switching times | -- | ≥2 million times | Self-made on/off tester |

PDLC film is driven by AC power, colorless and transparent when power is added, and fogged and translucent when power is removed, and its photoelectric performance is shown in the table above.

## Product Transparency / Haze

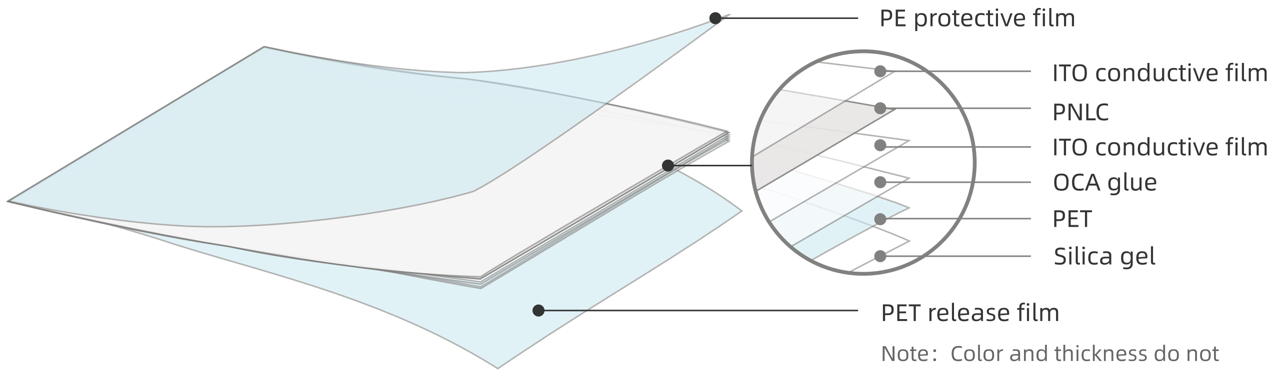
FILMBASE Warrior

| Items | Full light tester | | | | Parallel light tester | |
|---|---|---|---|---|---|---|
| | ON/OFF | Haze | Full light transmittance | b value | 45° light | Parallel light |
| / | OFF | 94.5 | 78.0 | 5.93 | / | / |
| | ON | 4.53 | 83.5 | 3.77 | 58.0 | 81.0 |




# FILMBASE Reverse

Filmbase Reverse film is a special transparent liquid crystal film, can be sandwiched in  the glass or adhesive on the glass, become opaque when power on , can be widely used in space partition, electronic products, automotive fields ,can be sandwiched into the glass or Adhesive onto the exited glass.




Note: Color and thickness do not represent the actual situation, only for the convenience of illustration.

## Size

FILMBASE Reverse

| Items | | | |
|---|---|---|---|
| Adhesive PNLC film | Total Thickness | | 0.37mm |
| | Cut to Sheet | Width | 100-1400(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1200,1400(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |
| Laminated PNLC film | Total Thickness | | 0.28mm |
| | Cut to Sheet | Width | 100-1400(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1200,1400(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |

## Optical Parameter

FILMBASE Reverse

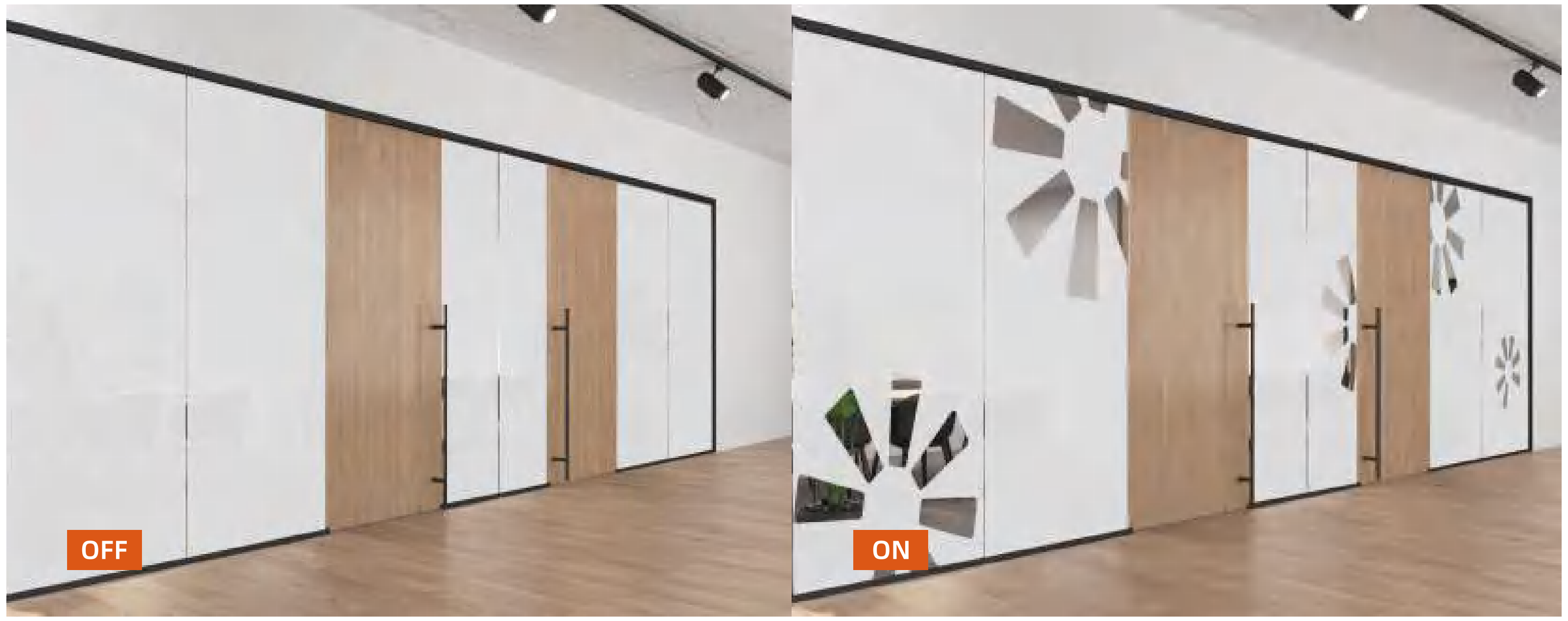
| Items | ON/OFF | | Test Methods/Standards |
|---|---|---|---|
| Operating voltage | ON | 60V | Multimeter/ Voltage Regulator,WGW |
| Power consumption | ON | agv 4-6/$m^2$ | Multi-parameter electric measuring instrument |
| Viewing angle | OFF | ≥ 160°+1 | Visual Inspection |
| Uv blocking rate | ON | ≥ 85% | Optical transmittance tester |
| Infrared blocking rate | ON | ≥ 60% | |
| Response time | OFF→ON | ≤ 200 ms | Liquid crystal parameter integrated tester |
| | ON→OFF | ≤ 220 ms | |
| Operating temperature | -- | -30°C ~ 50°C | constant temperature and humidity testing machine |
| Life-span | -- | 100000 Hours | GB18910.5-2008 |
| Switching times | -- | ≥2 million times | Self-made on/off tester |

PNLC film is driven by AC power,  fogged and translucent when power is added, colorless and transparent when power is removed, and its photoelectric performance is shown in the table above.

## Product Transparency / Haze

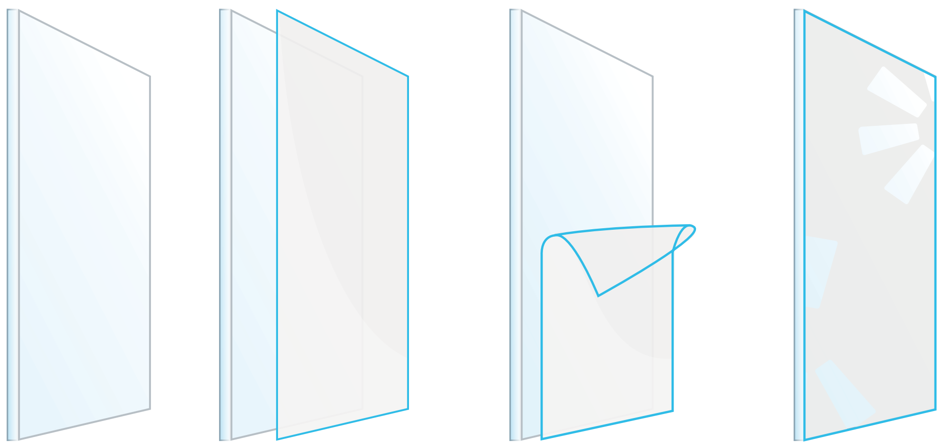
FILMBASE Reverse

| Items | Full light tester | | | | Parallel light tester | |
|---|---|---|---|---|---|---|
| | ON/OFF | Haze | Full light transmittance | b value | 45° light | Parallel light |
| Adhesive PNLC film | OFF | 7 | 90.0 | 3.85 | 80.0 | 85.0 |
| | ON | 92 | 88.0 | 4.42 | / | 19.0 |
| Laminated PNLC film | OFF | 7 | 90.0 | 1.71 | 80.0 | 85.0 |
| | ON | 92 | 88.0 | 1.84 | / | 19.0 |




# FILMBASE Limited edition

Limited edition is Filmbase luxury customized series, which allow etching content inside the smart film , such as company logo, slog,products, art paintings, pictures etc. , All contents can be switchable automatically. Can be sandwiched into the glass or Adhesive onto the exited glass to click to privacy.



## Size

FILMBASE Limited edition

| Items | | | |
|---|---|---|---|
| Adhesive PDLC film | Total Thickness | | 0.47mm |
| | Cut to Sheet | Width | less than1500 |
| | | Length | less than1500 |
| Laminated PDLC film | Total Thickness | | 0.38mm |
| | Cut to Sheet | Width | less than1500 |
| | | Length | less than1500 |

## Optical Parameter

FILMBASE Limited edition

| Items | ON/OFF | | Test Methods/Standards |
|---|---|---|---|
| Operating voltage | ON | 60V | Multimeter/ Voltage Regulator,WGW |
| Power consumption | ON | ≤ 5W/m² | Multi-parameter electric measuring instrument |
| Viewing angle | ON | ≥ 160° | Visual Inspection |
| Uv blocking rate | OFF | ≥ 99% | Optical transmittance tester |
| Infrared blocking rate | OFF | ≥ 60% | |
| Response time | OFF→ON | ≤ 30 ms | Liquid crystal parameter integrated tester |
| | ON→OFF | ≤ 200 ms | |
| Operating temperature | -- | -30℃ ~ 50℃ | constant temperature and humidity testing machine |
| Life-span | -- | 100000 Hours | GB18910.5-2008 |
| Switching times | -- | ≥2 million times | Self-made on/off tester |

## Product Transparency / Haze

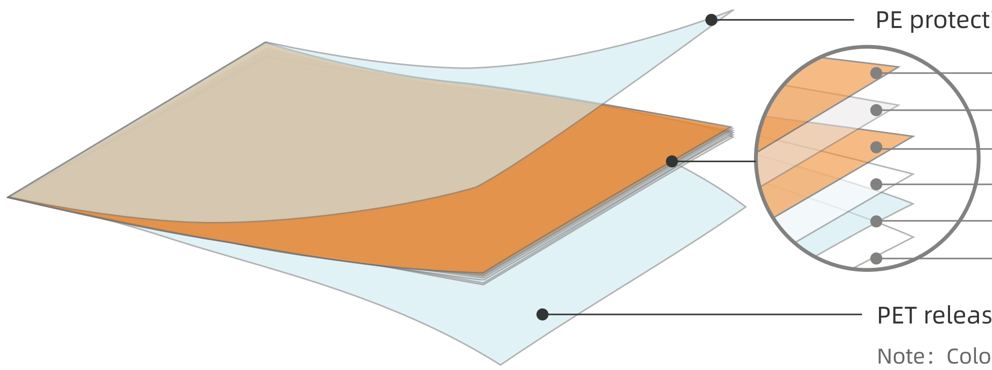
FILMBASE Limited edition

| Items | Full light tester | | | | Parallel light tester | |
|---|---|---|---|---|---|---|
| | ON/OFF | Haze | Full light transmittance | b value | 45° light | Parallel light |
| Adhesive PDLC film | OFF | 94.5 | 78.0 | 6.95 | / | / |
| | ON | 4.53 | 88.0 | 4.71 | 58.0 | 86.0 |
| Laminated PDLC film | OFF | 94.5 | 78.0 | 5.9 | / | / |
| | ON | 4.53 | 88.0 | 3.72 | 58.0 | 86.0 |

增加防刮系列单独页面

# FILMBASE Colorful

Filmbase Colorful PDLC film is designed for the application where need different colors such as gray, black, red, green, blue or other colors. Warmly welcomed by art centers, kindergartens, supermarkets, theme restaurants, places that need diversified design.can be sandwiched into the glass or Adhesive onto the exited glass to click to privacy.




Note: Color and thickness do not represent the actual situation, only for the convenience of illustration.

## Size

**FILMBASE Colorful**

| Items | | | |
|---|---|---|---|
| Gray adhesive PDLC film | Total Thickness | | 0.47mm |
| | Cut to Sheet | Width | 100-1800(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1200,1500,1800(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |
| Gray laminated PDLC film | Total Thickness | | 0.38mm |
| | Cut to Sheet | Width | 100-1800(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1200,1500,1800(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |
| Other colors adhesive PDLC film (black,blue,green,orange,red) | Total Thickness | | 0.54mm |
| | Cut to Sheet | Width | 100-1500(mm) |
| | | Length | 100-50000(mm) |
| | Roll | Width | 1500(mm) |
| | | Length | 30meters /50meters |
| | Roll tube | Inner Diameter | 6 inches |

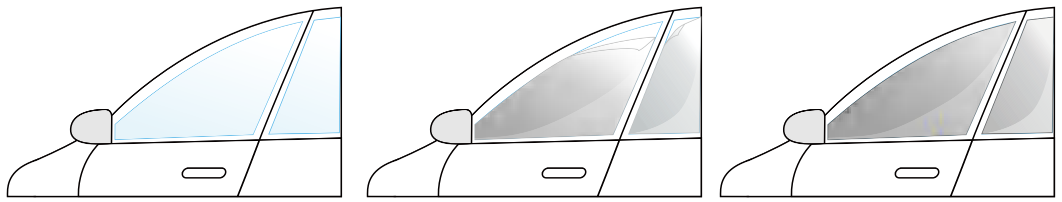
## Optical Parameter

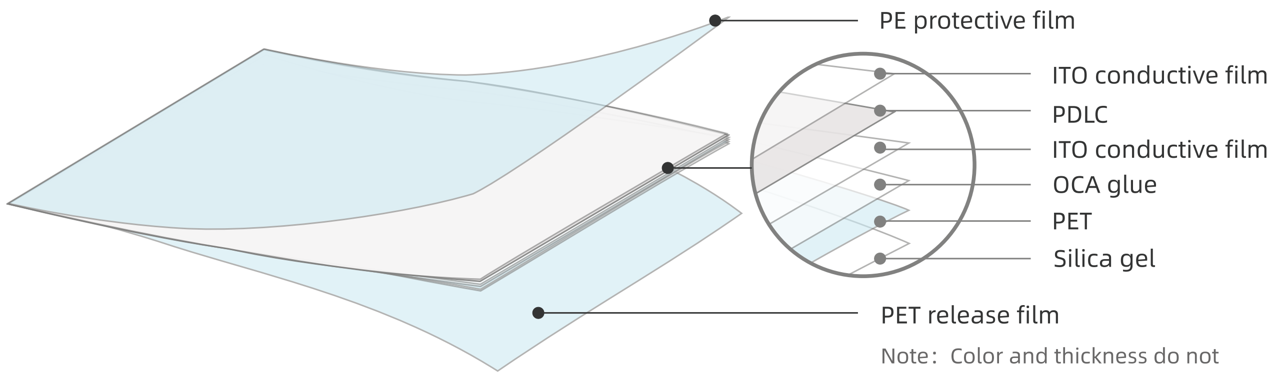
**FILMBASE Colorful**

| Items | ON/OFF | | Test Methods/Standards |
|---|---|---|---|
| Operating voltage | ON | 60V | Multimeter/ Voltage Regulator,WGW |
| Power consumption | ON | ≤ 5W/m² | Multi-parameter electric measuring instrument |
| Viewing angle | ON | ≥ 160° | Visual Inspection |
| Uv blocking rate | OFF | ≥ 99% | Optical transmittance tester |
| Infrared blocking rate | OFF | ≥ 60% | |
| Response time | OFF→ON | ≤ 30 ms | Liquid crystal parameter integrated tester |
| | ON→OFF | ≤ 200 ms | |
| Operating temperature | -- | -30°C ~ 50°C | constant temperature and humidity testing machine |
| Life-span | -- | 100000 Hours | GB18910.5-2008 |
| Switching times | -- | ≥2 million times | Self-made on/off tester |

# FILMBASE Auto Smart Film

Auto smart film is a smart film product developed by our company on the basis of the first generation of smart film, using special substrates and wide temperature liquid crystal materials, and specially developed for automotive use conditions.






Note: Color and thickness do not represent the actual situation, only for the convenience of illustration.

## Product Size

Auto smart film

| Items | | |
|---|---|---|
| Thickness | Total thickness (without protective film) | 0.13mm |
| | PDLC | 0.12mm |
| | Silica gel | 0.01mm |
| | PE protective film | 130μm |
| | PET release film | 50μm |
| Sheet | Width | 1200,1500[1]*mm |
| | Length | According to customer requirements |
| Roll | Width | 1250,1500[2]*mm |
| | Length | ≤ 50m/Roll |
| Roll tube | Inner Diameter | 6 inches |

Note*: This width is the actual size, the width of the calculated area is 1200,1500, mm

## Photoelectric Performance

Auto smart film

| Items | ON/OFF | | Test Methods/Standards |
|---|---|---|---|
| Operating voltage | ON | 60V | Multimeter/ Voltage Regulator,WGW |
| Power consumption | ON | ≤ 5W/m² | Multi-parameter electric measuring instrument |
| Viewing angle | ON | ≥ 160° | Visual Inspection |
| Uv blocking rate | OFF | ≥ 99% | Optical transmittance tester |
| Infrared blocking rate | OFF | ≥ 82% | |
| Response time | OFF→ON | ≤ 30ms | Liquid crystal parameter integrated tester |
| | ON→OFF | ≤ 200ms | |
| Operating temperature | -- | -30℃ ~ 95℃ | constant temperature and humidity testing machine |
| Lifetime | -- | 10 -15 years | GB18910.5-2008 |
| Switching times | -- | ≥ 100 ten thousand Times | Self-made on/off tester |

Auto smart filmis driven by AC power, gray and transparent when power is added, and fogged and translucent when power is removed, and its photoelectric performance is shown in the table above.

## Product Transparency / Haze

Auto smart film

| Items | Full light tester | | | | Parallel light tester | |
|---|---|---|---|---|---|---|
| | ON/OFF | Haze | Full light transmittance | b value | 45° light | Parallel light |
| Adhesive PDLC film | OFF | 94.2 | 51.19 | 5.24 | / | / |
| | ON | 2.94 | 56.43 | 3.2 | 38.4 | 55.26 |